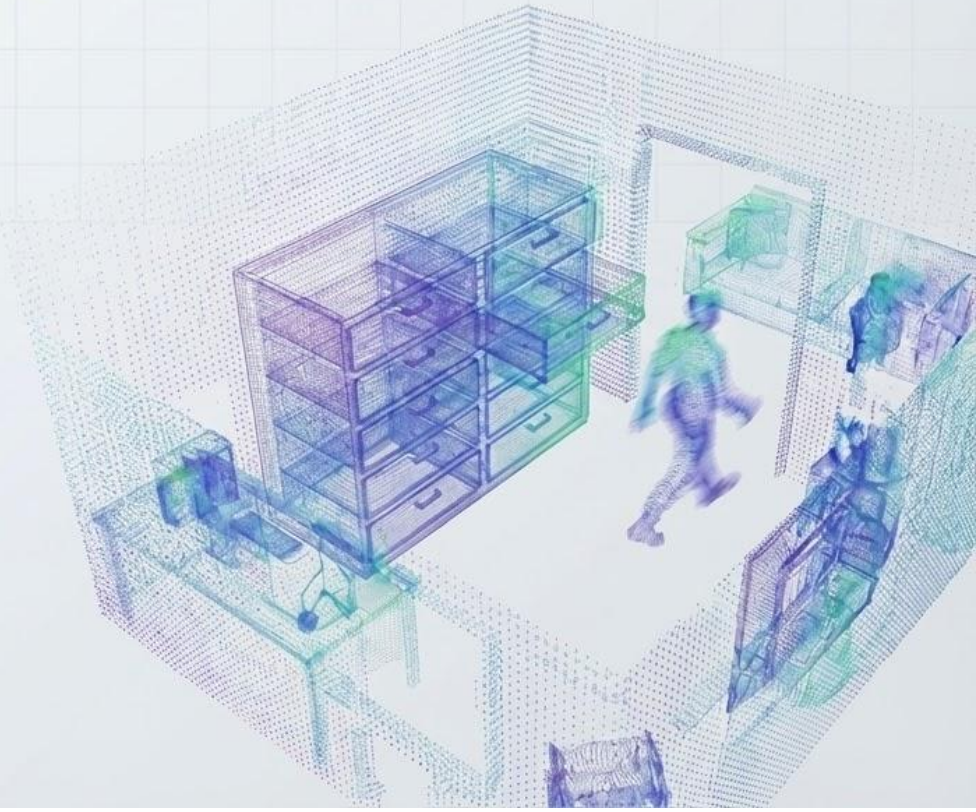
Mapping Quasi Dynamic Environments

(Cabinet with drawers)

Team 5

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- Ronit Shetty
- Gowtham Parasuraman
- Sairam Sridharan



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Why map Quasi-Dynamic Environments ?

Classical SLAM Assumes:

- landmarks do not move
- geometry is fixed across frames
- the robot's pose is the only unknown changing variable

In Quasi-Dynamic Environments:

- Drawer geometry **changes with configuration**
- Previously seen surfaces **appear/disappear**
- **Occlusions change** constantly
- Landmarks cannot be assumed persistent



This invalidates the core SLAM formulation, making optimization unstable.

Data Collection - Cabinet with Drawers in Highbay

Static Camera Drawers Move

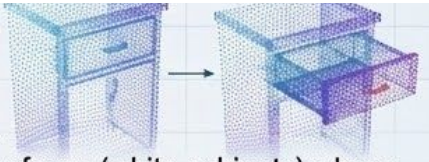




Camera and Drawers move

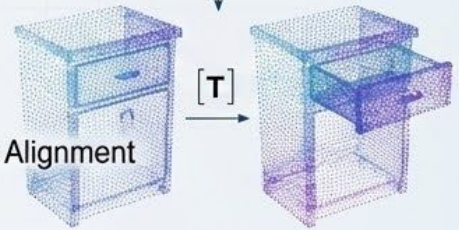
Methodology

1. Multi-State Reconstruction (SfM + MVS)



- **Concept:** Treat each articulation state (e.g., "Drawer 1 Open") as a distinct static reconstruction problem.
- **Robustness:** Utilize **SuperPoint** (features extractes) and **SuperGlue** (matching) to handle texture-less surfaces (white cabinets) where traditional SIFT fails.
- **Output:** Dense point clouds for Base (P_base {base}) and each Drawer state (P_n_{n}).

2. Global Geometric Alignment



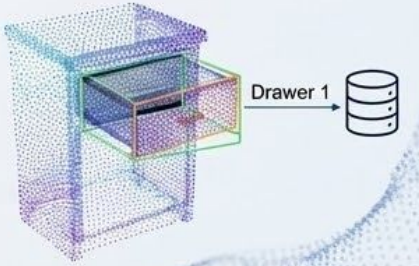
- **Objective:** Unify all states into a single coordinate system.
- **Method:** Downsample → FPFH descriptors → Compute FPFH descriptors $ → → → RANSAC Coarse Alignment
- **Critical Output:** Transformation matrix $\mathbf{T_{n \to base}}$ {\to base}$ linking State n to the Base.

3. Difference-Based Segmentation



- **Technique:** Spatial subtraction ($P_{n} – P_{base},$) with distance thresholding and clustering to remove outliers and add the individual drawers as prismatic joinst
- **Result:** Isolates the dynamic component (drawer geometry) → from the static cabinet body.

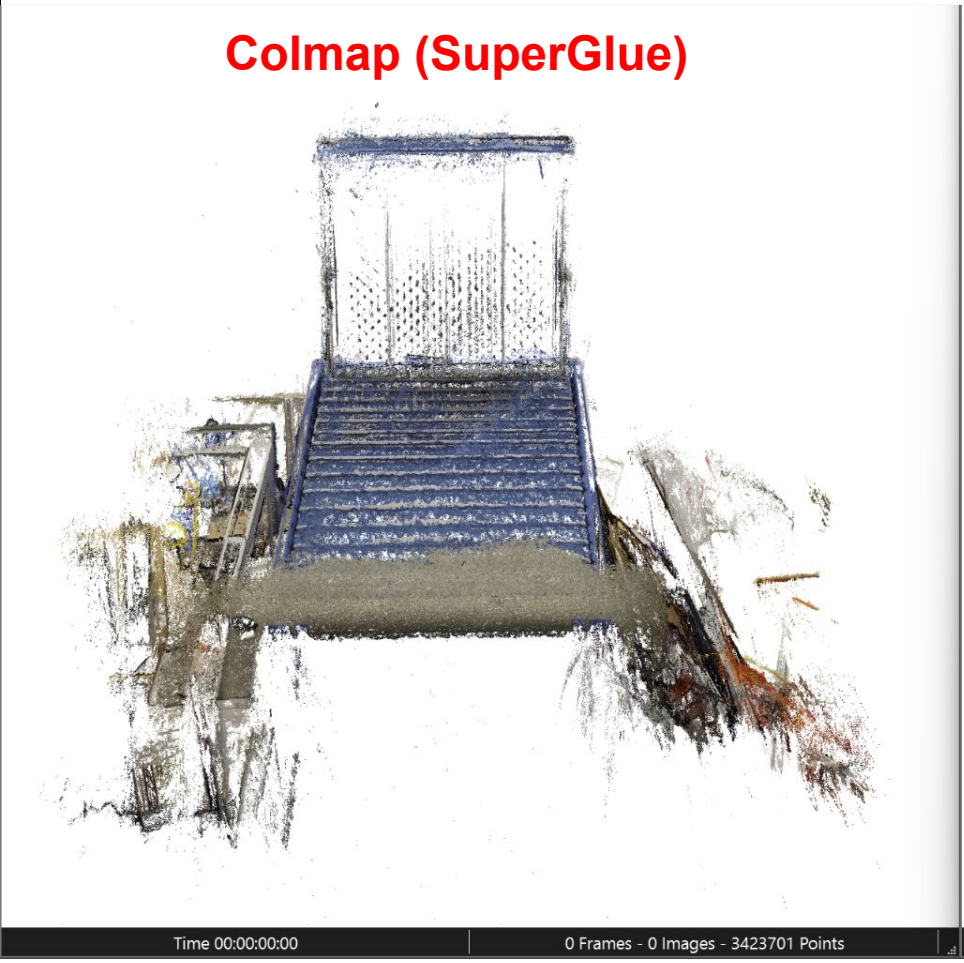
4. Semantic Injection & Mapping



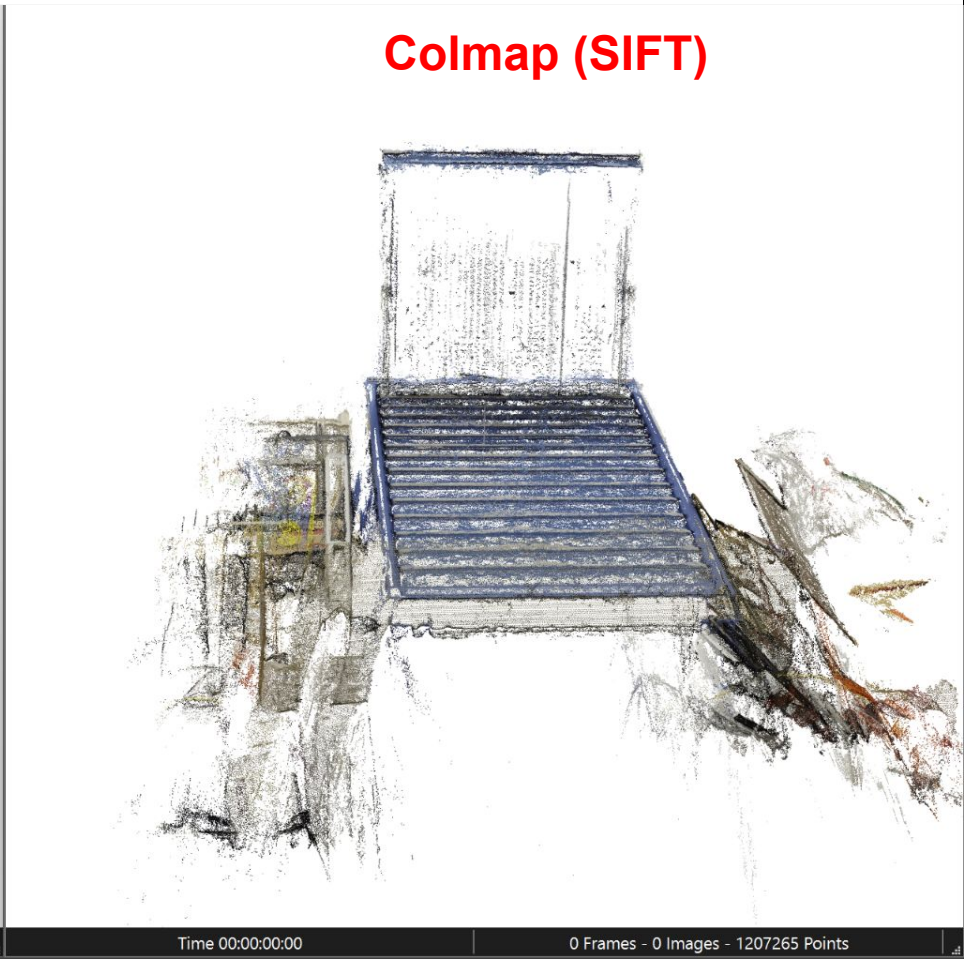
- **Process:** 2D Object Detection (GroundingDINO) + Depth Estimation.
- **Transform:** Back-project 2D to 3D, then apply alignment (p_{local}), then apply the alignment matrix: $$p_{global} = T_{n \to base} \cdot p_{local}$$
- **Storage:** Objects mapped to specific drawer IDs in a queryable SQL database

Comparison of Reconstruction

Colmap (SuperGlue)

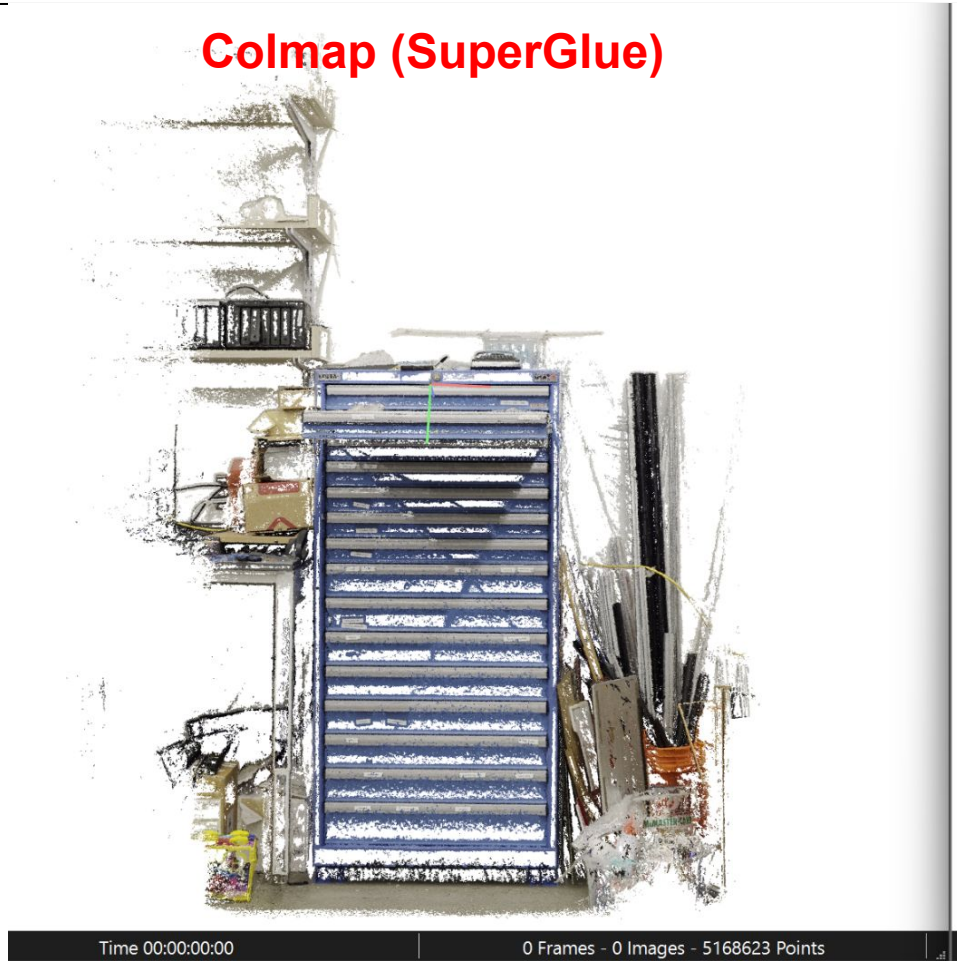


Colmap (SIFT)



Comparison of Reconstruction

Colmap (SuperGlue)

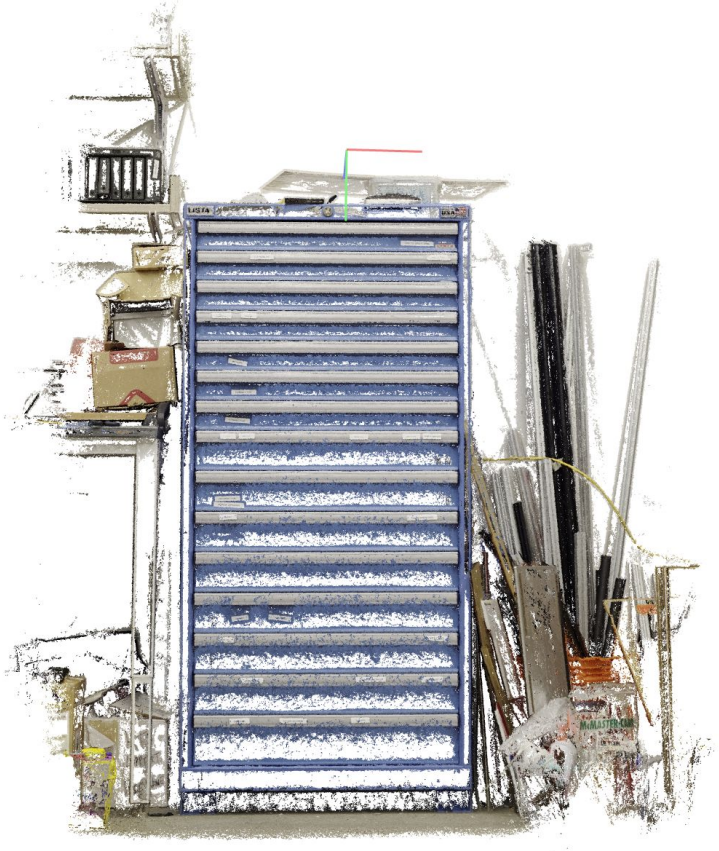


Colmap (SIFT)

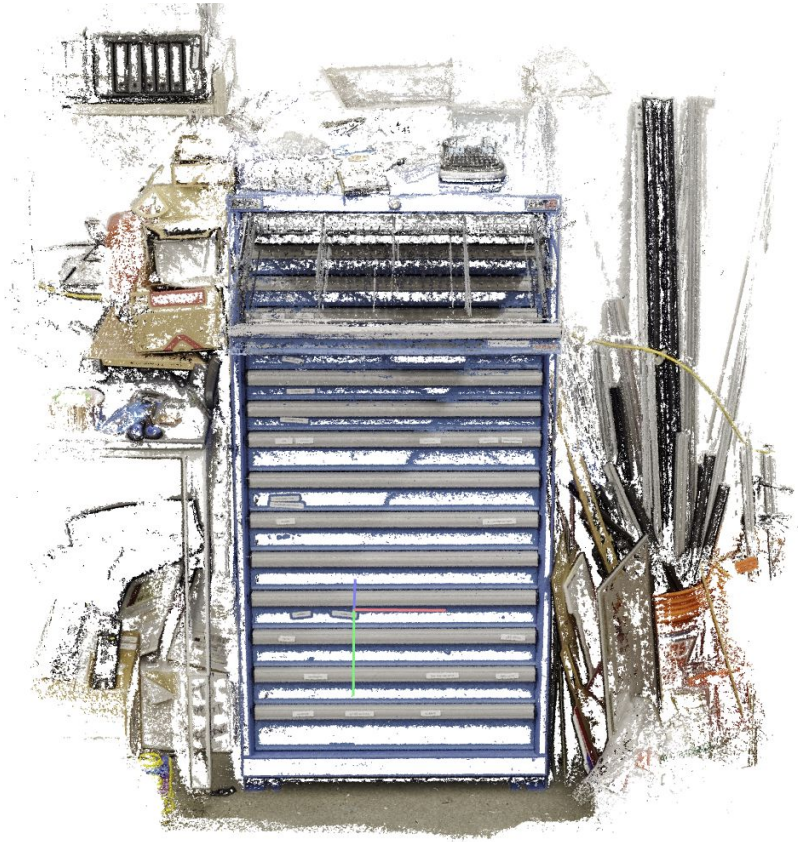


Reconstruction of All Drawers

Base



Drawer 1



Reconstruction of All Drawers

Drawer 2



Drawer 3

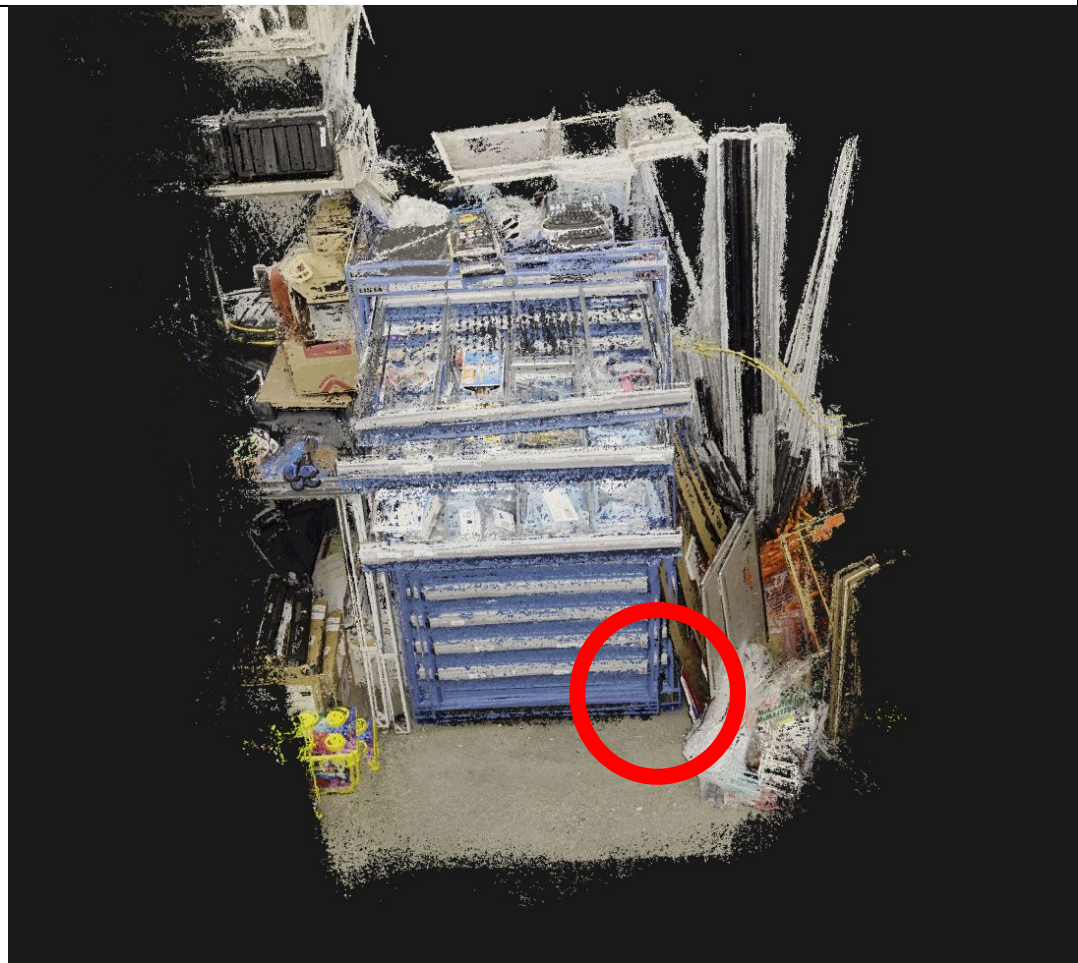


Why Alignment?

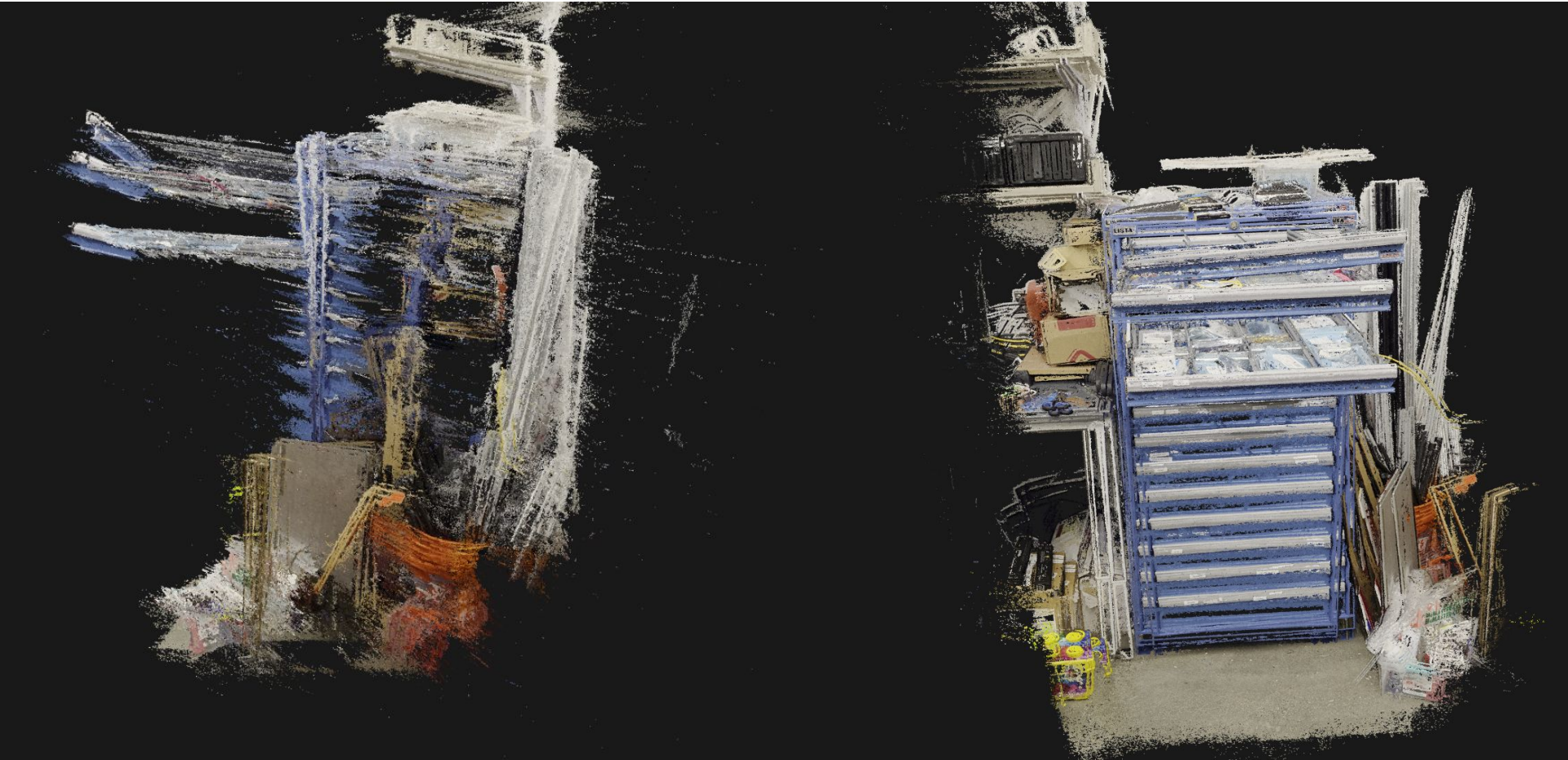
- **Base reconstruction** → Coordinate frame A (origin, scale, orientation)
- **Drawer 1 open** → Coordinate frame B (different origin, possibly different scale)
- **Drawer 2 open** → Coordinate frame C (yet another frame)

Alignment Method:

- **RANSAC + FPFH** registration
- Registration helps with:
  - Compute meaningful differences
  - Merge into a coherent model
  - Track articulated parts

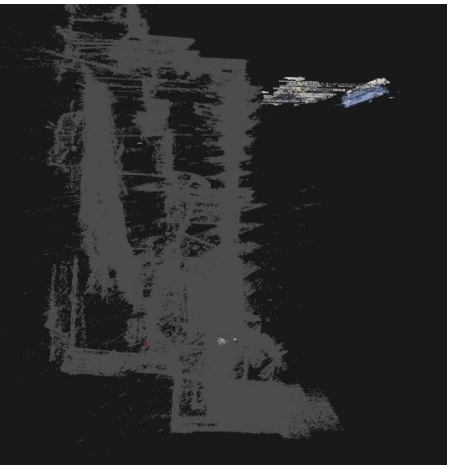


Alignment (Other View)

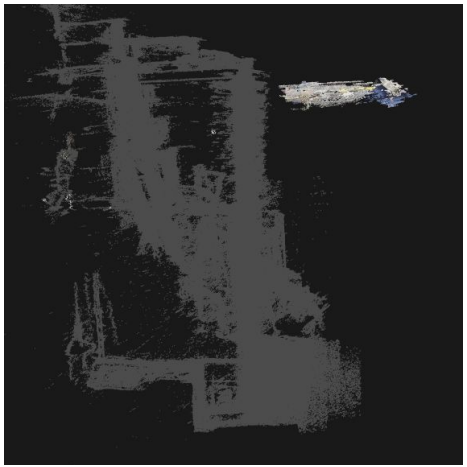


Segmentation

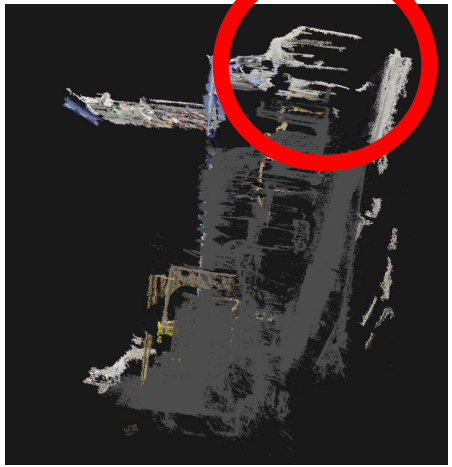
Drawer 1



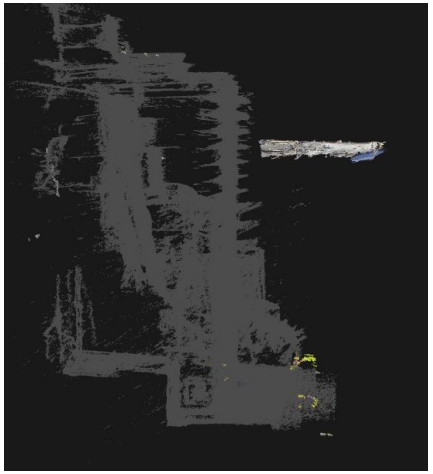
Drawer 2



Drawer 3

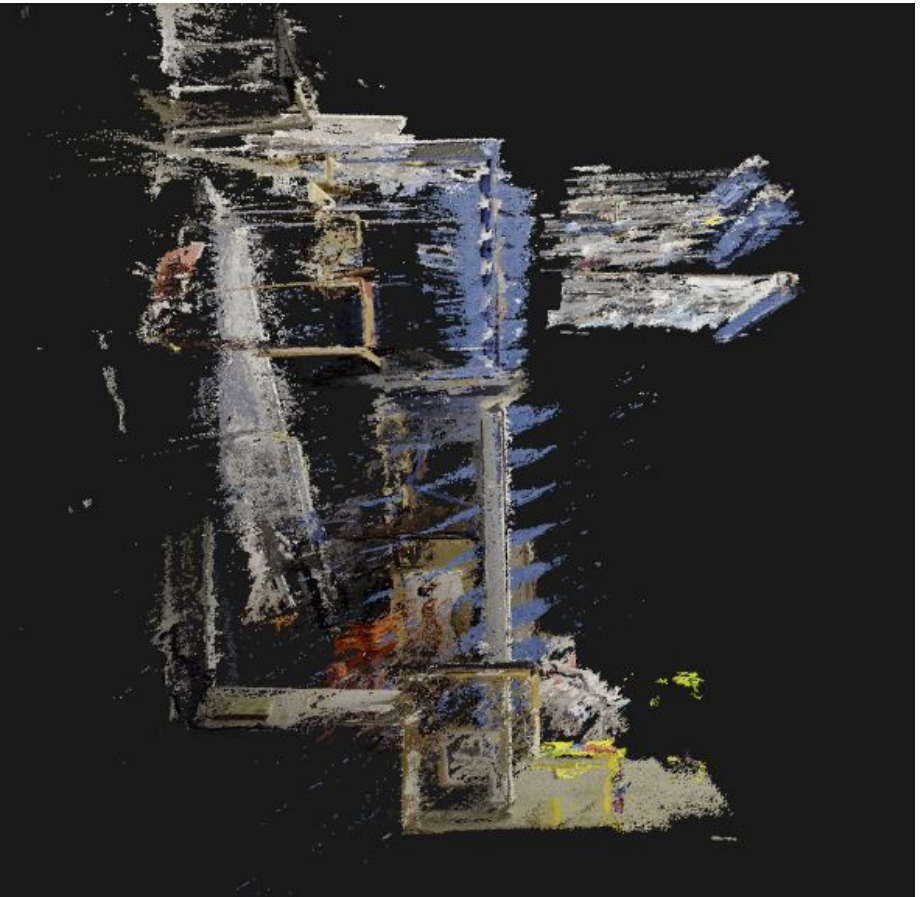


Drawer 4



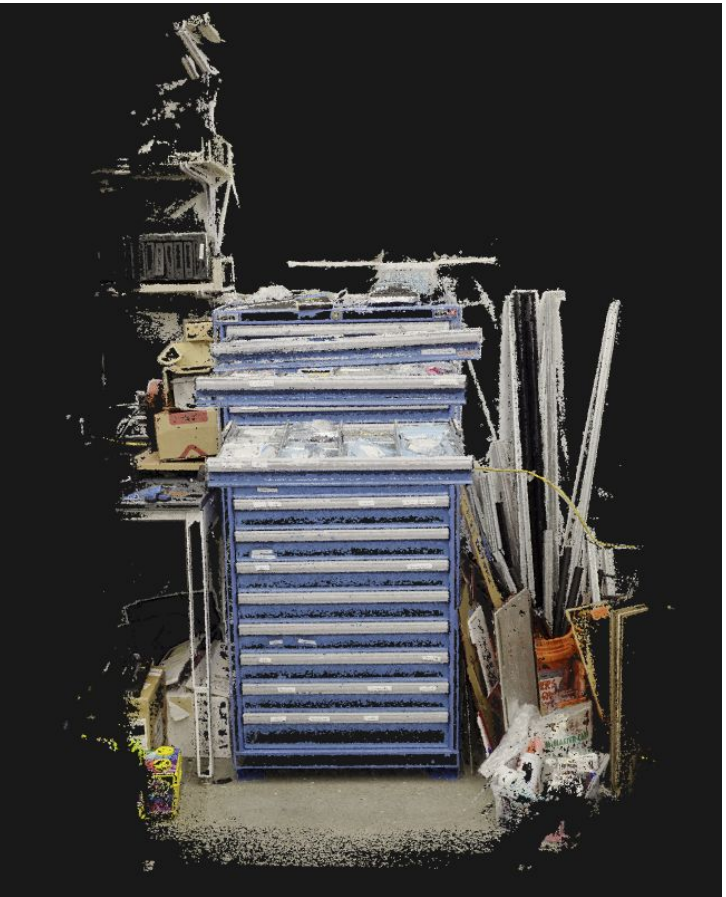
- Remove Base reconstruction

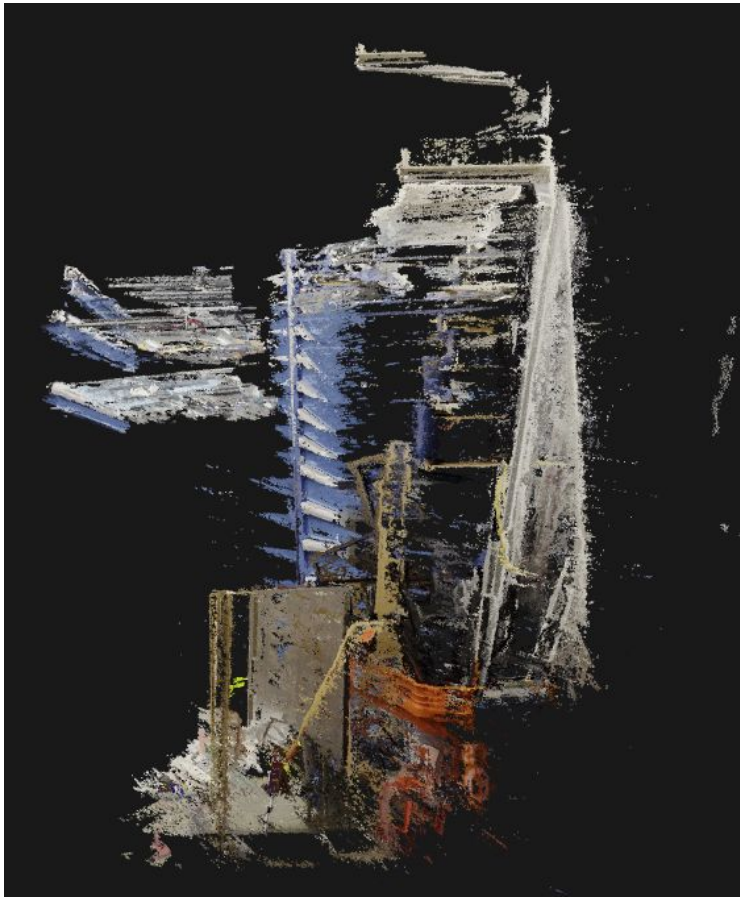
Merged after Extraction (Open and Close States)



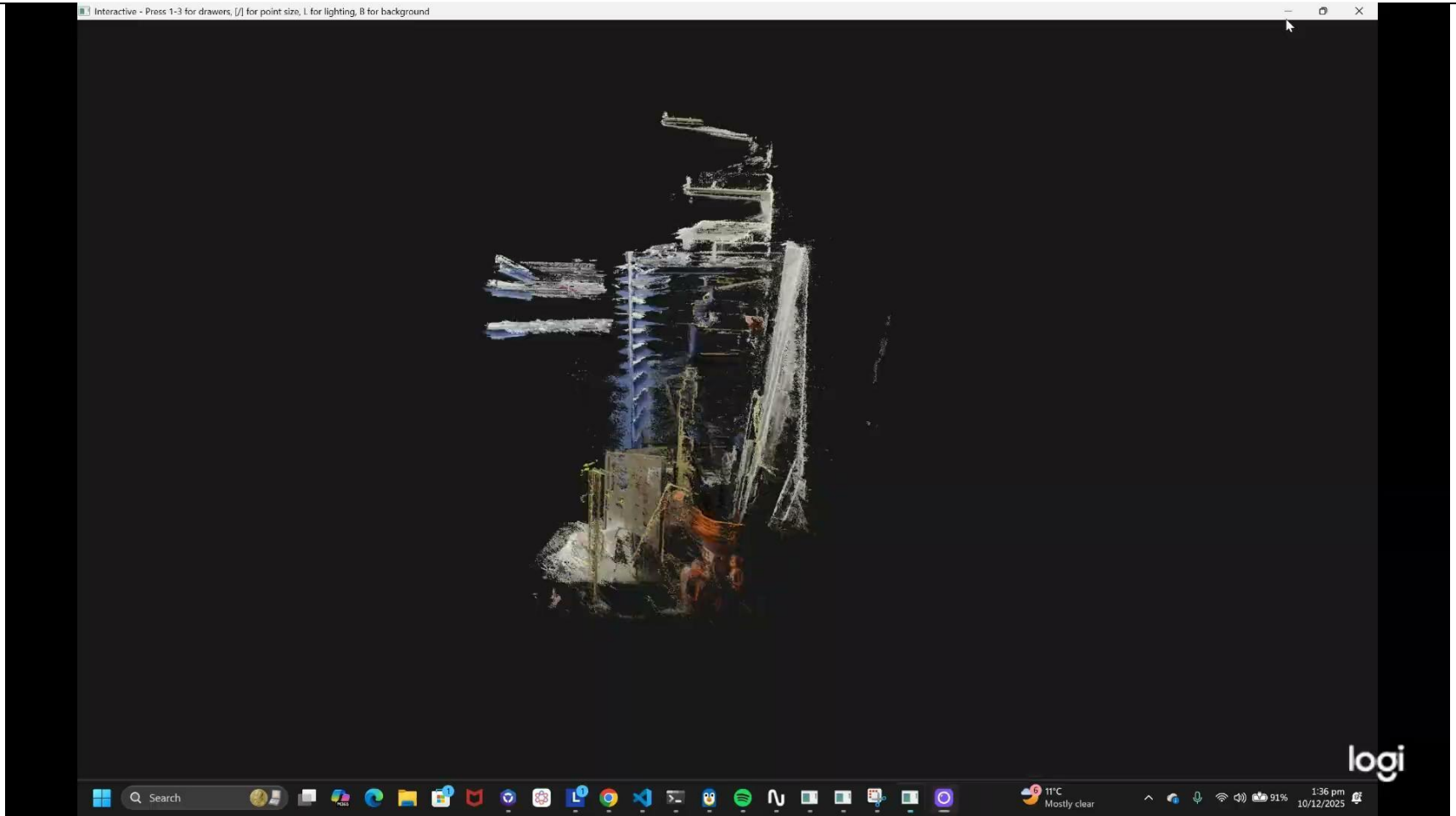


Merged after Extraction

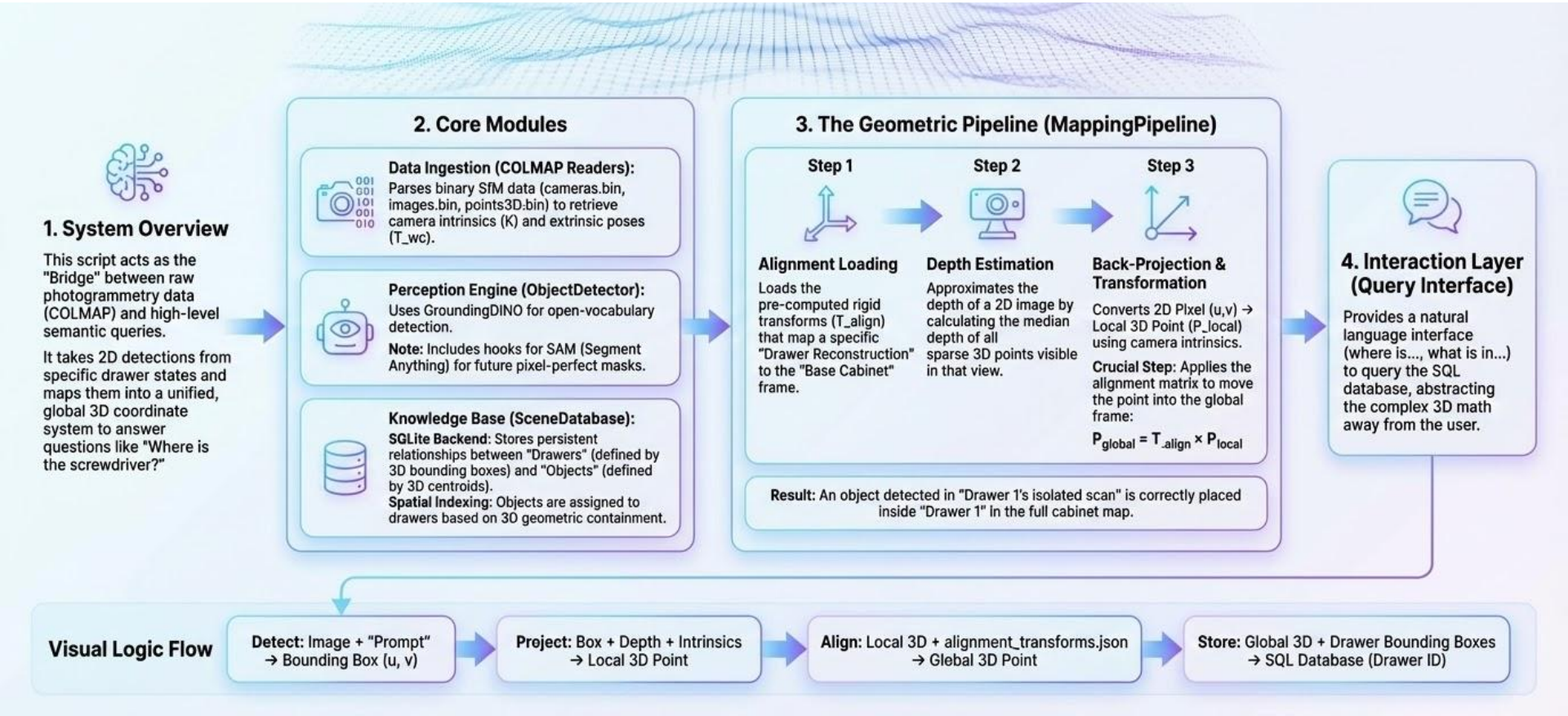




Articulation



Object Mapping



Limitations of Our Methodology

- Drawers were modeled at Dynamic, What about rest of the scene ?
- Scale Estimation - Point Cloud Matching
- Model the ‘not so perfect’ joints
  - How to ideally model a prismatic joint ?
- After Disturbance?
  - Each Scene has to be remapped - Needs Movement ??
  - Why not robot searches the drawers one by one and queries each image ??

Learning Based Methods

- Articulate Anything: https://github.com/vlongle/articulate-anything
- DRAWER: https://github.com/xiahongchi/DRAWER/tree/main
- IAAO: https://github.com/lulusindazc/IAAO

Articulate Anything (Demo)

